

Farhan Akhtar Makandar

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer with 1+ year of research experience specializing in building production-ready ML systems using PyTorch, TensorFlow, and scikit-learn. Proven track record of achieving 22% accuracy improvement over traditional ML baselines through Temporal Fusion Transformer architecture. Strong foundation in statistical modeling, feature engineering, data pipelines, and model deployment. Passionate about bridging the gap between research prototypes and scalable ML systems.

TECHNICAL SKILLS

- Programming: Python, SQL, R, C
- ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost
- Deep Learning: CNN, RNN, LSTM, GANs, Transformer Architectures
- Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering
- MLOps: Git/GitHub, Model Validation, Hyperparameter Tuning, Experiment Tracking
- Cloud & Tools: AWS, Oracle Cloud, MongoDB, Power BI, DAX
- Concepts: Statistical Modeling, Cross-Validation, Ensemble Methods, Time Series Analysis, EDA

PROFESSIONAL EXPERIENCE

Research Intern - AI/ML - NIT Calicut

Jun 2024 - Dec 2024

- Developed and evaluated transformer-based models for complex predictive tasks using PyTorch, achieving rigorous performance benchmarks
- Collaborated on cutting-edge AI research focused on neural network optimization and deep learning applications for engineering problems
- Conducted comprehensive data analysis, feature engineering, and model validation using scientific computing libraries
- Assisted in documenting research findings and preparing technical reports for publication

PROJECTS

Seismic Response Prediction

Built Temporal Fusion Transformer (TFT) model achieving 22% improvement over XGBoost and KNN baselines for seismic structural response prediction. Performed exploratory data analysis using Pandas, NumPy, and Seaborn. Implemented hyperparameter optimization and rigorous model evaluation pipeline.

Techn

SAINT Model - Tabular Classification

Implemented Self-Attention and Intersample Attention (SAINT) mechanism for tabular data classification. Optimized feature interaction learning through attention-based architecture on structured datasets.

Techn

Sales Analytics Dashboard

Designed Power BI dashboard with advanced DAX formulas for real-time sales analysis, KPI tracking, customer segmentation, and business intelligence insights for stakeholder reporting.

Techn

EDUCATION

B.E. in Artificial Intelligence & Machine Learning

Yenepoya Institute of Technology, Moodbidri (VTU) | Expected 2026

CERTIFICATIONS

- IBM Python for Data Science (Credly Verified)
- Machine Learning with Python - IBM Cognitive Class
- Deep Learning Fundamentals - IBM Cognitive Class
- Accelerating Deep Learning with GPUs - IBM Cognitive Class
- Supervised Machine Learning - Coursera
- Building RAG Agents with LLMs - NVIDIA
- AWS Generative AI Essentials - AWS
- AWS Prompt Engineering - AWS
- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- Accenture Data Analytics Simulation - Forage
- MongoDB Atlas - MongoDB University
- HTML Training - IIT Bombay